

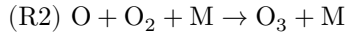
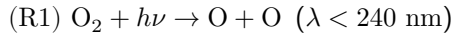
7 Stratospheric Ozone - Chapman Mechanism

Stratospheric Chemistry

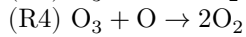
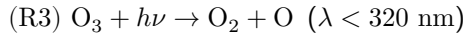
- Most UV-A (320nm – 400nm) reaches the surface.
- Ozone screens out most UV-B (280nm – 320nm).
- UV-C (180nm – 280nm) can't reach the surface.

Chapman Mechanism

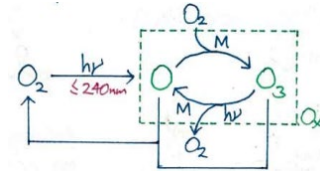
- Ozone formation



- Ozone destruction



- Lifetime of the O atom



$$\tau_O = \frac{[O]}{k_2[O][O_2][M]} = \frac{1}{k_2[O_2][M]} \approx 1 \text{ s}$$

- Odd oxygen family $[O_x] = [O_3] + [O]$, assuming both $[O]$ and $[O_3]$ are in PSSA.

$$\frac{d[O_x]}{dt} = 2k_1[O_2] - 2k_4[O_3][O] = 0 \text{ (slow and exactly balance)}$$

$$k_2[O][O_2][M] = k_3[O_3] \Rightarrow \frac{[O]}{[O_3]} = \frac{k_3}{k_2[O_2][M]} = \frac{k_3}{k_2 C_{O_2} n_a^2} = \frac{\tau_O}{\tau_{O_3}} \ll 1$$

- Lifetime of O_x

$$\tau_{O_x} = \frac{[O_x]}{2k_4[O_3][O]} = \frac{1}{2k_4[O]}$$

- Concentration of O_x

$$[O_3] \sim \left(\frac{k_1 k_2}{k_3 k_4} \right)^{1/2} C_{O_2} n_a^{3/2}$$

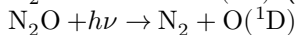
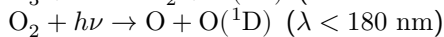
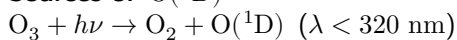
- Overestimates the observed O_3 concentrations by a factor of 2 or more. There must be additional sinks for O_3 not accounted for by the Chapman mechanism.

Atomic Oxygen



8 electrons with two unpaired valence shell electrons in its “triplet” $O(^3P)$ ground state ($2s^2 2p_x^2 2p_y^2 2p_z^2$) and is therefore a biradical. Atomic oxygen with all electrons paired ($2s^2 2p_x^2 2p_y^2$) is in higher energy “singlet” $O(^1D)$ is more reactive.

Sources of $O(^1D)$



Loss of $O(^1D)$

- $O(^1D) + M \rightarrow O + M^*$

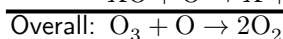
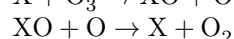
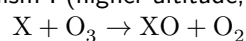
PSSA of $O(^1D)$

$$J_{O_3}[O_3] = k_{O(^1D)+M}[O(^1D)][M] \Rightarrow [O(^1D)] = \frac{J_{O_3}[O_3]}{k_{O(^1D)+M}[M]}$$

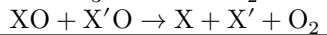
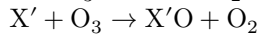
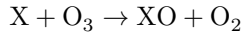
8 Stratospheric Ozone – Catalytic Destruction Cycles

Catalytic Ozone Destruction Mechanisms

- Mechanism I (higher altitude, abundant oxygen atom)



- Mechanism II (lower altitude, oxygen atom does not participate)

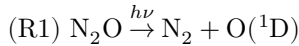


Overall: $2O_3 \rightarrow 3O_2$

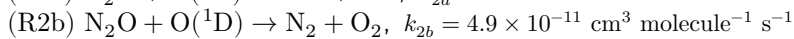
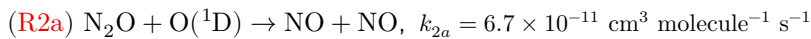
- X is regenerated in the process as a catalyst.

NO_x Cycle

- $NO_x = NO + NO_2$, $NO_y = NO + NO_2 + HNO_3 + N_2O_5$.
- Stratospheric source of NO_x is from N_2O , and oxidation of atmospheric N_2 at the high temperatures of the aircraft engine.
- Photolytic destruction of N_2O

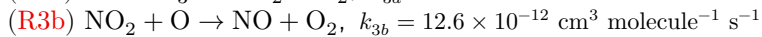
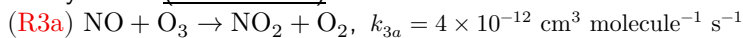


- Chemical loss of N_2O and Yield of NO_x



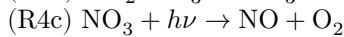
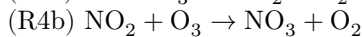
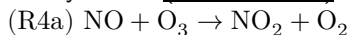
$NO \text{ yield} = (2k_{2a}[O(^1D)]) / (J_{N_2O} + (k_{2a} + k_{2b})[O(^1D)])$, 11% at 30 km

- Catalytic loss (Mechanism I)

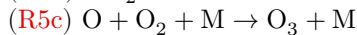
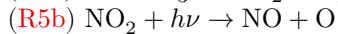
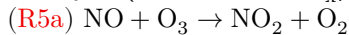


Fast conversion between NO and NO_2 : $\tau_{NO} = 1/k_{3a}[O_3] \approx 70 \text{ s}$, $\tau_{NO_2} = 1/k_{3b}[O_3] \approx 800 \text{ s}$, $[O] \approx 10^8 \text{ molecule cm}^{-3}$

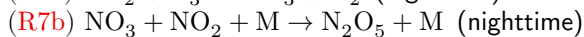
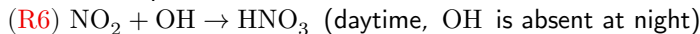
- Catalytic loss (Mechanism II)



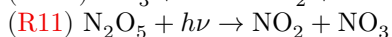
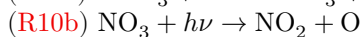
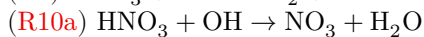
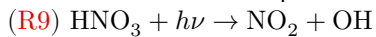
Null cycle (No net loss of O_x)



- Termination processes



- Conversion of reservoir species to NO_x , lifetime of HNO_3 : weeks, lifetime of N_2O_5 : hours to days



Ultimate removal of NO_3 is by transport to the troposphere where HNO_3 is rapidly removed by deposition.

- Rate limiting step

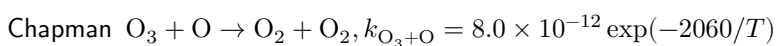
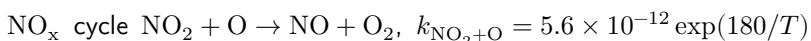
Consider the catalytic loss cycle and the null cycle:

$$0 = \frac{d[NO_2]}{dt} \approx k_{3a}[NO][O_3] - J_{NO_2}[NO_2] - k_{3b}[NO_2][O]$$

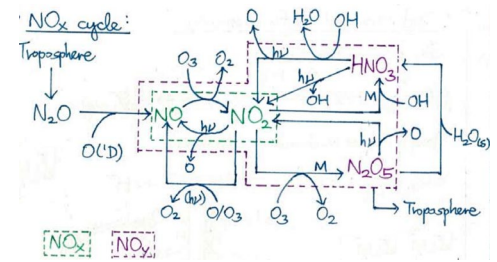
$$-\frac{d[O_3]}{dt} \approx -\frac{d[O_x]}{dt} = k_{3a}[NO][O_3] + k_{3b}[NO_2][O] - J_{NO_2}[NO_2] = 2k_{3b}[NO_2][O]$$

From (R3a), one O_3 lost; From (R3c), one O atom lost. Rate limiting step is (R3b), the slower of the two branches would lead to loss. In regions where sunlight is strong (sunlit stratosphere), photolysis is fast, so the null cycle dominates. However, in certain conditions (e.g., polar winter/spring, or when odd oxygen atoms are abundant), the reaction with O can become significant.

- NO_x cycle involve O atom is more effective than Chapman Mechanism in upper stratosphere.
- Ratio of the loss rates

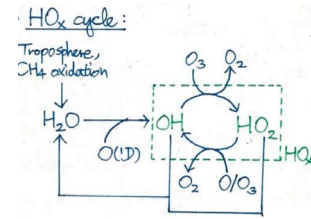


$$\frac{\text{Loss by } NO_x}{\text{Loss by Chapman}} = \frac{k_{NO_2+O}[NO_2][O]}{2k_{O_3+O}[O_3][O]} = 4.5 \text{ (at 35 km)}$$



HO_x Cycle

- $\text{HO}_x = \bullet\text{OH} + \text{HO}_2 \bullet$
- HO_x production
 - (R1) $\text{H}_2\text{O} + \text{O}(^1\text{D}) \rightarrow 2\text{OH}$ ($\text{H}_2\text{O} \sim 5\text{ppm}$ from troposphere / methane oxidation, 90% of HO_x production)
 - (R2) $\text{CH}_4 + \text{O}(^1\text{D}) \rightarrow \text{OH} + \text{CH}_3$ ($\text{CH}_4 \sim 0.5\text{ppm}$, 10% of HO_x production)
- Catalytic loss
 - (R3) $\text{OH} + \text{O}_3 \rightarrow \text{HO}_2 + \text{O}_2$
 - (R4a) $\text{HO}_2 + \text{O} \rightarrow \text{OH} + \text{O}_2$ (Mechanism I, high altitude)
 - (R4b) $\text{HO}_2 + \text{O}_3 \rightarrow \text{OH} + 2\text{O}_2$ (Mechanism II, low altitude)
- Null cycle
 - (R5a) $\text{HO}_2 + \text{NO} \rightarrow \text{OH} + \text{NO}_2$
 - (R5b) $\text{NO}_2 + h\nu \rightarrow \text{NO} + \text{O}$
- Terminating step
 - (R6) $\text{OH} + \text{HO}_2 \rightarrow \text{H}_2\text{O} + \text{O}_2$
 - (R7) $\text{OH} + \text{NO}_2 \rightarrow \text{HNO}_3$
- Odd oxygen loss rate



$$\frac{d[\text{O}_3]}{dt} = -k_3[\text{OH}][\text{O}_3] - k_{4b}[\text{HO}_2][\text{O}_3], \frac{d[\text{O}]}{dt} = -k_{4a}[\text{HO}_2][\text{O}] + J_{5b}[\text{NO}_2]$$

$$k_3[\text{OH}][\text{O}_3] = k_{5a}[\text{HO}_2][\text{NO}] + k_{4a}[\text{HO}_2][\text{O}] + k_{4b}[\text{HO}_2][\text{O}_3]$$

$$J_{5b}[\text{NO}_2] = k_{5a}[\text{HO}_2][\text{NO}]$$

$$\frac{d[\text{O}_x]}{dt} = -2k_{4b}[\text{HO}_2][\text{O}_3] - 2k_{4a}[\text{HO}_2][\text{O}]$$

The loss is governed by two cycles, with their significance depending on the altitude.

- Effect of NO_x

At 20 km, $k_{\text{OH}+\text{O}_3}[\text{OH}][\text{O}_3] = k_{\text{HO}_2+\text{NO}}[\text{HO}_2][\text{NO}] \Rightarrow \frac{[\text{HO}_2]}{[\text{OH}]} = \frac{k_{\text{OH}+\text{O}_3}[\text{O}_3]}{k_{\text{HO}_2+\text{NO}}[\text{NO}]} \approx 4.4$. 80% of HO_x exists as HO_2 .

Raising the concentration of NO_x reduces the catalytic loss of O_x .

- Comparison with Chapman Mechanism

At 45 km, $\frac{\text{Loss by HO}_x}{\text{Loss by Chapman}} = \frac{k_{\text{HO}_2+\text{O}}[\text{HO}_2][\text{O}]}{2k_{\text{O}_3+\text{O}}[\text{O}_3][\text{O}]} = 0.4$, only 1/2 as effective as compared to Chapman.

ClO_x Cycle

- $\text{ClO}_x = \text{Cl} \bullet + \text{ClO} \bullet$, $\text{Cl}_y = \text{Cl} + \text{ClO} + \text{HOCl} + \text{ClONO}_2 + \text{HCl}$
- CFC (chlorofluorocarbons) has no significant sink in the troposphere – removed mainly by OH radicals (slow). Lifetime of CFCl_3 (CFC-11): 45 years; Lifetime of CF_2Cl_2 (CFC-12): 100 years
- Total organic chlorine peaked at ~ 3.7 ppb between mid-1992 and mid-1994, slowly declining in the global troposphere due to the Montreal protocol and amendments leading to CFC ban.
- Cl production
 - (R1) $\text{CFCl}_3 + h\nu \rightarrow \text{CFCl}_2 + \text{Cl}$ ($\lambda < 240$ nm)
 - (R2) $\text{CF}_2\text{Cl}_2 + h\nu \rightarrow \text{CF}_2\text{Cl} + \text{Cl}$ ($\lambda < 240$ nm)
- Catalytic loss
 - (R3) $\text{Cl} + \text{O}_3 \rightarrow \text{ClO} + \text{O}_2$
 - (R4) $\text{ClO} + \text{O} \rightarrow \text{Cl} + \text{O}_2$ (Mechanism I, high altitude)

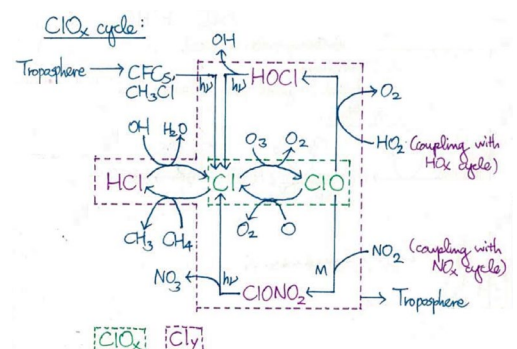
$$\tau_{\text{Cl}} = \frac{[\text{Cl}]}{k_{\text{Cl}+\text{O}_3}[\text{Cl}][\text{O}_3]} = 0.2 \text{ s}, \tau_{\text{ClO}} = \frac{[\text{ClO}]}{k_{\text{ClO}+\text{O}}[\text{ClO}][\text{O}]} = 53 \text{ s}, \text{ fast conversion}$$

Coupling with HO_x cycle (Mechanism II, lower altitude)

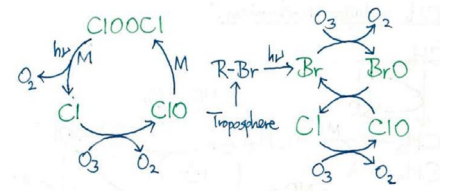
- (R5a) $\text{OH} + \text{O}_3 \rightarrow \text{HO}_2 + \text{O}_2$
- (R5b) $\text{ClO} + \text{HO}_2 \rightarrow \text{HOCl} + \text{O}_2$
- (R5c) $\text{HOCl} + h\nu \rightarrow \text{OH} + \text{Cl}$

Self reaction of ClO (Mechanism II, lower altitude)

- (R6a) $2 \times (\text{Cl} + \text{O}_3 \rightarrow \text{ClO} + \text{O}_2)$
- (R6b) $2\text{ClO} \rightarrow \text{ClOOCl}$
- (R6c) $\text{ClOOCl} + h\nu \rightarrow \text{ClO} + \text{ClO}$
- (R6d) $\text{ClO} + h\nu \rightarrow \text{Cl} + \text{O}_2$



- Null cycle
(R7a) $\text{Cl} + \text{O}_3 \rightarrow \text{ClO} + \text{O}_2$
(R7b) $\text{ClO} + \text{NO} \rightarrow \text{Cl} + \text{NO}_2$
(R7c) $\text{NO}_2 + h\nu \rightarrow \text{NO} + \text{O}$
- Terminating steps
(R8) $\text{CH}_4 + \text{Cl} \rightarrow \text{HCl} + \text{CH}_3$
(R9) $\text{ClO} + \text{NO}_2 \rightarrow \text{ClONO}_2$
- Conversion to reactive species
(R10) $\text{HCl} + \text{OH} \rightarrow \text{Cl} + \text{H}_2\text{O}$
(R11a) $\text{ClONO}_2 + h\nu \rightarrow \text{ClO} + \text{NO}_2$
(R11b) $\text{ClONO}_2 + h\nu \rightarrow \text{Cl} + \text{NO}_3$
- Loss Rate



$$\frac{d[\text{O}_x]}{dt} = -2k_4[\text{ClO}][\text{O}]$$

$$k_3[\text{Cl}][\text{O}_3] = k_4[\text{ClO}][\text{O}] + k_{7b}[\text{ClO}][\text{NO}] \Rightarrow \frac{[\text{Cl}]}{[\text{ClO}]} = \frac{k_4[\text{O}] + k_{7b}[\text{NO}]}{k_3[\text{O}_3]} \sim 0.008 \text{ (at 40 km), most of the } \text{ClO}_x \text{ are in } [\text{ClO}].$$

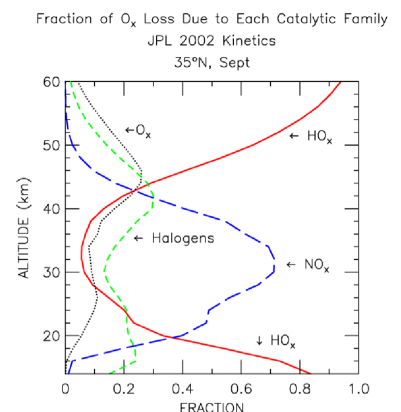
- Effect of NO_x
At 45 km, $k_{\text{ClO}+\text{O}}[\text{ClO}][\text{O}] = k_{\text{ClO}+\text{NO}}[\text{ClO}][\text{NO}]$.
~50% ClO goes to null cycle, and ~50% of ClO participates in the catalytic cycle.
Raising the concentration of NO_x reduces the catalytic loss of O_x .
- Comparison with Chapman Mechanism

$$\text{At 45 km, } \frac{\text{Loss by } \text{ClO}_x}{\text{Loss by Chapman}} = \frac{k_{\text{ClO}+\text{O}}[\text{ClO}][\text{O}]}{2k_{\text{O}_3+\text{O}}[\text{O}_3][\text{O}]} = 1, \text{ roughly as effective as compared to Chapman.}$$

- BrO_x catalytic system
 $\text{Br} \sim 20 \text{ ppt}$ and $\text{Cl} \sim 3400 \text{ ppt}$, but O_3 destruction rate by BrO_x is at the same order as ClO_x .
Reason: $\text{Br}_y \rightarrow \text{Br}$ rapidly; Br_y mainly exists as $\text{Br} + \text{BrO}$ and not in the reservoir species; BrNO_3 unstable – can rapidly photolyse.

Fraction of O_x Loss due to Each Catalytic Family

- Below ~25 km, the major determinants of O_3 loss are HO_x and halogen chemistry, with NO_x playing a much smaller role.
- At altitudes from ~25 km to 38 km, NO_x cycles dominate among the various catalytic cycles.
- At altitudes > 38 km, HO_x , O_x , and ClO_x become more important.



9 Stratospheric Ozone – Ozone Hole Formation

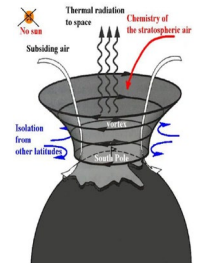
Polar Ozone Depletion Theories

Depletion is confined to the spring months (September to November), near-total in the lower stratosphere (approx. 15–20 km altitude).

1. Polar vortex in wintertime: isolates air, allows extreme cooling
2. Formation of polar stratospheric clouds (PSC): provides surfaces for reactions
3. Heterogeneous reactions (reactions on the surface of PSC): converts reservoir species to active Cl
4. Liberation of active Cl (normally tied up via bonding with ClO_x reservoir species such as ClONO_2 and HCl)
5. Removal of NO_x from the stratosphere by the sedimentation of PSC: suppress the reformation of reservoir species such as ClONO_2
6. Catalytic destruction of O_3 in the presence of sunlight in springtime

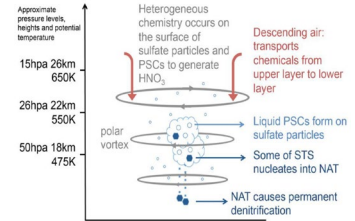
Polar Vortex

- Sunlight does not reach the south pole during winter polar night
- A strong, persistent, circumpolar wind field forms over Antarctica during the winter and early spring.
- Especially stable in Antarctic due to the symmetric distribution of ocean and ice around the continent.



Polar Stratospheric Cloud (PSC)

- When temperatures inside the polar vortex drop below $\sim -78^\circ\text{C}$ (193 K), the air becomes cold enough for clouds to form in the extremely dry stratosphere. Form at altitudes between 15 and 25 km.
- Composition:
 - Type I PSCs (NAT): Not water clouds. Primarily composed of nitric acid trihydrate ($\text{HNO}_3 \cdot 3\text{H}_2\text{O}$).
 - Type II PSCs: Composed of water ice, form at even colder temperatures below -88°C .

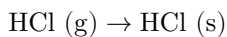


Heterogeneous Chemistry

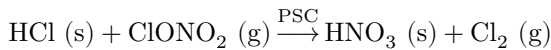
- PSC promotes the conversion of inactive Cl reservoir species to active Cl, by concentrating the reactant molecules.
- Formation of HNO_3 is assisted by hydrogen bonding to the water molecules in the PSC particles.

- Pathway 1: $\text{HCl (g)} \rightarrow \text{Cl}_2 \text{ (g)}$

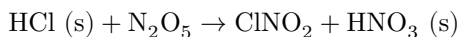
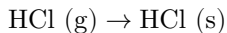
Absorption of gaseous HCl by PSCs



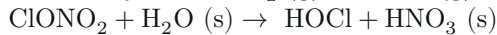
Reaction of gaseous ClONO_2 with HCl on the PSC particles



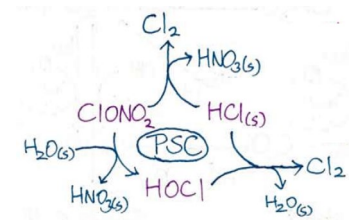
- Pathway 2: $\text{HCl (g)} \rightarrow \text{ClONO}_2 \text{ (g)}$ in presence of N_2O_5



- Pathway 3: $\text{ClONO}_2 \text{ (g)} \rightarrow \text{HOCl (g)}$

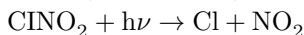
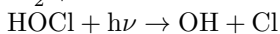
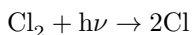


- $\text{N}_2\text{O}_5 + \text{H}_2\text{O (s)} \rightarrow 2\text{HNO}_3 \text{ (s)}$



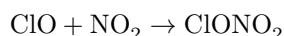
Liberation of Active Chlorine

- Active chlorine species: Cl_2 , HOCl , ClONO_2 , rapidly photolyzed when sunlight returns in spring, suddenly releasing a massive amount of Cl atoms into the lower stratosphere.



Denitrification

- Removal of gaseous N species
- Major process: formation of NAT PSCs; Minor process: Formation of HNO_3 from gaseous N species and subsequent retention of HNO_3 (s).
- PSCs particles can grow larger over the summer and sink to lower altitudes (sedimentation). This permanently removes nitrogen oxides (NO_x) from the air mass inside the vortex.
- If denitrification didn't happen, the HNO_3 produced could eventually photolyze ($\text{HNO}_3 + h\nu \rightarrow \text{OH} + \text{NO}_2$). The NO_2 produced would then react with ClO to reform the reservoir ClONO_2 , shutting down the ozone destruction cycle.



Catalytic Ozone Destruction (The Dimer Cycle)

In the polar lower stratosphere, there aren't many oxygen atoms because sunlight is still weak.

1. **Initiation:** $\text{Cl} + \text{O}_3 \rightarrow \text{ClO} + \text{O}_2$ (chlorine atom destroys one ozone, forming ClO).
2. **Dimer Formation:** $\text{ClO} + \text{ClO} \rightarrow \text{ClOOC}$ (two ClO molecules combine to form a chlorine peroxide dimer).
3. **Photolysis:** $\text{ClOOC} + h\nu \rightarrow \text{Cl} + \text{ClOO}$ (sunlight breaks the dimer).
4. **Decomposition:** $\text{ClOO} \rightarrow \text{Cl} + \text{O}_2$ (the ClOO radical falls apart).

Chronology of Antarctic Ozone Hole

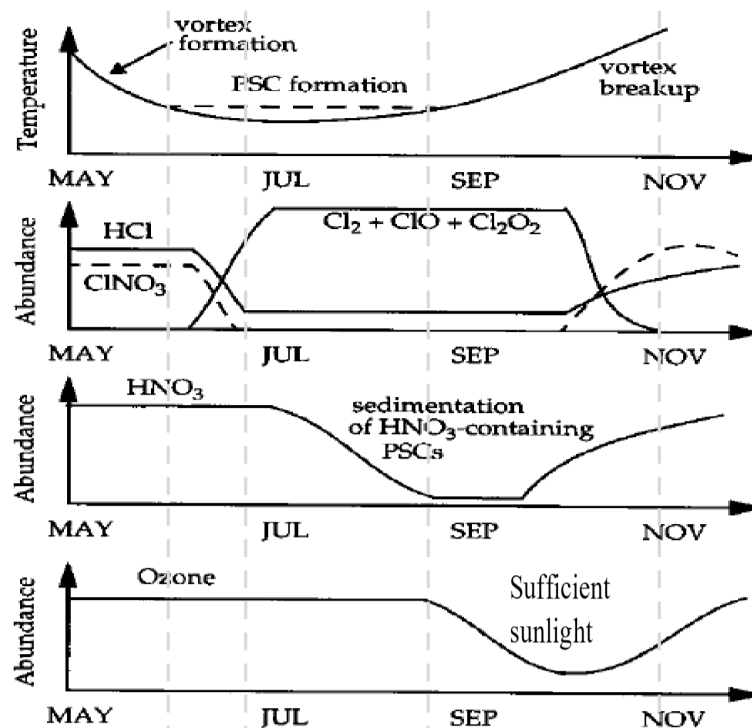


Fig. 10-13 Chronology of the antarctic ozone hole.

Season	Months	Polar Vortex	Sunlight	PSCs	Key Chemical Process	Ozone Level
Autumn	Mar–May	Forming	Decreasing	None	Reservoir buildup	Normal
Winter	Jun–Aug	Strong & Stable	None	Extensive	Heterogeneous activation on PSCs & Denitrification	Normal (but primed for loss)
Spring	Sep–Nov	Strong	Returning	Present	Photolysis of Cl_2 & Catalytic Dimer Cycle	Drops rapidly
Summer	Dec–Feb	Breaks down	High	None	Mixing with mid-latitude air	Recovers to normal

Antarctic and Arctic Ozone Hole

Feature	Antarctica	Arctic
Polar Vortex	Strong, stable, long-lasting	Weaker, more prone to disruption
Winter Temperatures	Very cold ($< -80^\circ\text{C}$)	Warmer by $\sim 10\text{K}$ on average
PSC Formation	Extensive, long-lived	Less frequent, shorter duration
Denitrification	Efficient (PSCs sediment out)	Inefficient (PSCs evaporate, releasing HNO_3)
Result	Large, predictable ozone hole each spring	High interannual variability; large depletion possible only in exceptionally cold years

- The 2011 Event: An unusually cold and persistent Arctic vortex led to ozone depletion approaching Antarctic levels, showing the potential risk if the Arctic continues to cool in the stratosphere.
- Polar ozone loss depends on temperature and active halogen concentrations in lower stratosphere.
- Ozone destruction in arctic is less dramatic compared with the Antarctic.
 1. Winter stratosphere of the Arctic is warmer by $\sim 10\text{K}$ on average.
 2. The warmer temperature results in less PSC and shorter presence time.
 3. The less abundant and less persistent PSC dramatically reduce the extent of denitrification.
 4. PSC in the Arctic does not have sufficient time to settle out of the stratosphere.
 5. PSC releases their HNO_3 back to the stratosphere, making ClOx polar cycle less efficient.

Montreal Protocol

- The agreement: signed in 1987, the Montreal Protocol on Substances that Deplete the Ozone Layer is an international treaty to phase out the production of CFCs, halons, and other ozone-depleting substances by 2000.
- Recover: Assuming continued compliance, the ozone layer is projected to recover to pre-1980 levels by the middle to late 21st century.